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Autunian age constrained by fold tests for paleomagnetic data from the Mezarif and Abadla basins (Algeria)

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Abstract

A paleomagnetic study has been conducted on a formation dated as Autunian in the Nekheila area (31.4°N, 1.5°W) in the Mezarif basin. ChRM was thermally isolated in 117 samples from seven sites. This ChRM ($D = 131.8°$, $I = 15.7°$, $k = 196$, $\alpha_{95} = 3.8°$ after dip correction; corresponding pole 29.3°S, 56.4°E) is very similar to that obtained in the neighboring Abadla basin from a formation of the same age. Fold tests associated with progressive unfolding applied to the full merged data from the dated formations of these two basins clearly indicate that the magnetization acquisition predates the deformation, which is attributed to the last phase of the late-Hercynian. The magnetization in these basins is therefore primary or acquired just after deposition. For the African Apparent Polar Wander Path, the age of the paleomagnetic poles of the Autunian part is now confirmed by paleomagnetic test.
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Keywords: Africa; Paleomagnetism; Autunian; Apparent Polar Wander Path; Fold test; Rotation

1. Introduction

The main uncertainties in the African Apparent Polar Wander Path (APWP) for the Upper Paleozoic, are related to the lack of suitable formations on the cratonic areas for paleomagnetic studies and the insufficient control of the age of magnetization acquisition. In fact, the age of most paleomagnetic results are not constrained by any paleomagnetic test as they have been acquired for tabular formations with no conglomerate or intrusive rocks. Several such studies were performed in the Reggane (Daly and Irving, 1983), Illizi (Henry et al., 1992; Derder et al., 1994, 2001a,b; Kies et al., 1995), Abadla (Morel et al., 1981; Merabet et al., 1998) and Tindouf (Merabet et al., 1999; Henry et al., 1999) basins within the Saharan craton, but until recently, the only argument for their age of magnetization acquisition was the fact that their paleomagnetic direction differed according to the age of the formation. Consequently, none of these formations had been affected by the same magnetic overprint event. Positive reversal tests were also been found in the Jurassic formations of the Illizi basin (Kies et al., 1995), but there were no proofs of the magnetization primary character. Henry et al. (2004) also obtained positive reversal tests for remagnetization of Upper Cenozoic age in several formations in the Illizi basin.

In Upper Carboniferous formations, Derder et al. (2001c,d) obtained positive fold test in two different areas of the Saharan Craton. Two Upper Carboniferous paleomagnetic poles have now well-established ages and have similar pole location as the other Upper Carboniferous formations. The aim of this paper is to constrain, by fold tests, the age of the paleomagnetic data of the African Autunian formations. Autunian age formations with slightly different dip are exposed on the sides of the large Zousfana anticline south of Bechar. Paleomagnetic works were carried out in the western side, in the Abadla basin (Morel et al., 1981; Merabet et al., 1998) and this study covers the basin on the eastern side of this anticline, the Mezarif basin.

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2. Geology and sampling

The large Zousfana anticline is complex one with an average NNE–SSW to N–S trend. Its exposed stratigraphic units are mainly Carboniferous limestones, which form the Djebel Béchar and Djebel Mezarif on the two limbs (Fig. 1). The two basins of Abadla and Mezarif are separated by this anticline. They are partially covered by the Cenozoic Hamada lacustrine limestone and small ergs. In the Mezarif basin 70 km east of Bechar town, only two depressions have been opened by the erosion expose Carboniferous and Permian units. In the Guern er Rechoua depression, the Autunian sandstones are coarse and unsuitable for paleomagnetic analysis.

The 6 km long Nekheila depression (31.4°N, 1.5°W), corresponds to a minor NE–SW anticline (Figs. 1 and 2(a)). Only its southeastern limb shows outcrops of Permian units, dipping here toward the SE to S. These units are disconformably overlain by Cenozoic limestone. Lithologically, this area comprises carbonate, sandstone, clay and pelites series. The alluvial material is made of red pelites with fine-grained sandstone intercalations. It is similar to that of the Abadla formation (of which the lower unit is dated as Autunian—Doubinger and Fabre, 1983) in the Bechar basin (Nedjari, 1982, 1990). Paleontologically, Deleau (1951) attributed a Stephanian age to the nekheila red formation. Nedjari (1982) found that the pollens, characteristic of the gray Autunian formations in the Permian European basins, occur at the base of this red pelitic formation. These deposits correspond to a tropical climate and show well preserved ferric material (Freytet in Nedjari, 1982). All the strata are perfectly concordant from Lower Carboniferous units to continental Autunian red deposits. The major Hercynian phase occurred in this area just after the Autunian, as in the Béchar one (Conrad and Le Mosquet, 1984). The Zousfana anticline area remains tectonically stable since the Hercynian phase (Deleau, 1951). Suitable outcrops were only available in the southern limb of the Nekheila anticline at few locations. This limb bends to the southeast to south with a dip between 10° and 25°. The sampling was restricted to seven sites (Fig. 2(a)) with oriented blocks collected in red pelitic beds throughout the thickness of the Autunian unit (Fig. 2(b)). A total of 68 large blocks, oriented using plaster cap technique, have been collected.

3. Laboratory and analytical procedures

3.1. Samples preparation

Laboratory analyses were undertaken in the Saint Maur laboratory. One or two cores, between 3 and 7 cm long and 2.5 cm in diameter, were drilled perpendicularly to the bedding from each oriented blocks. In most cases, each one of the 117 oriented cores was sliced into more than one cylindrical specimen which have 2.2 cm in height. When two samples were retained for paleomagnetic study from a same oriented block, they always have been taken from two distant cores and correspond to slightly different levels.

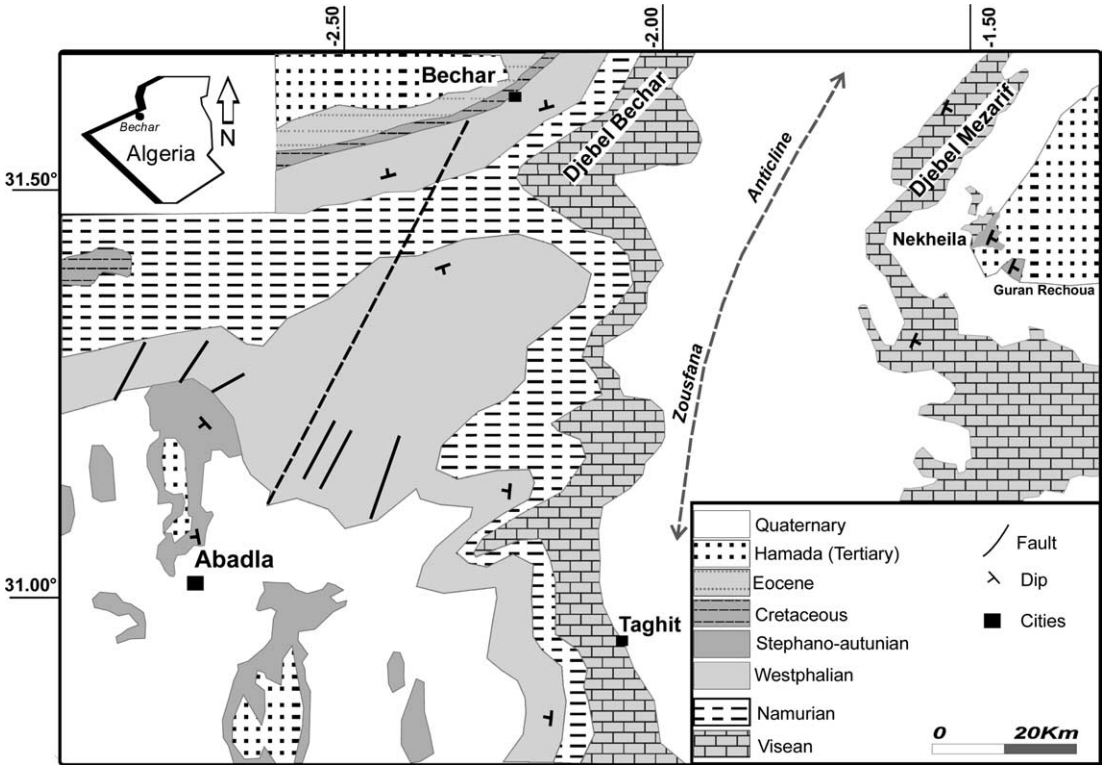


Fig. 1. Geological map of Bechar–Abadla–Mezarif area modified from Deleau (1951).

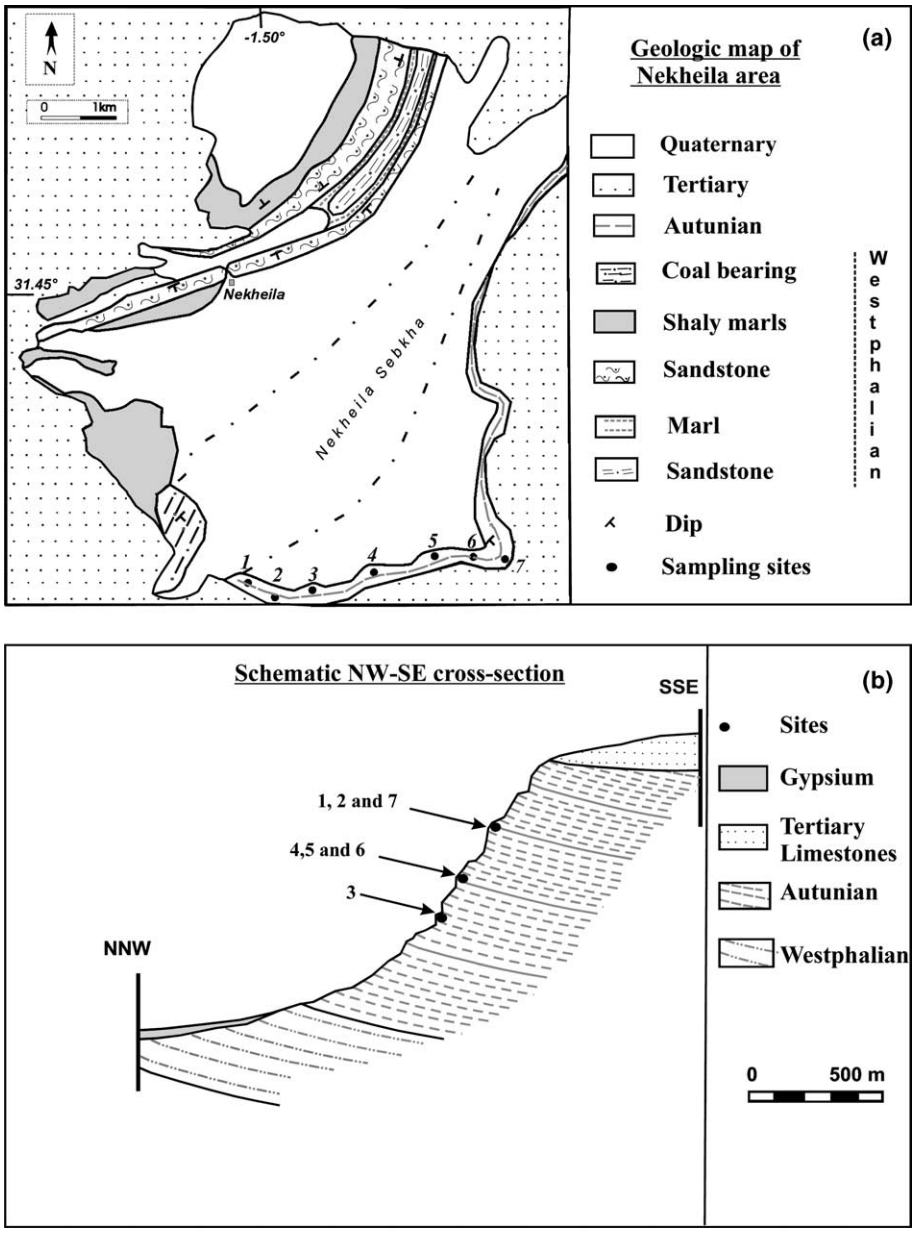


Fig. 2. (a) Geological map of Nekheila Hollow modified from Deleau (1951) and location of sampled sites of the Autunian red formation. (b) Schematic NW–SE cross-section of the Nekheila series and levels of the Autunian sampled sites.

3.2. Rock magnetic procedures and properties

The identification of the main ferrimagnetic carriers as well as the monitoring of their stability upon demagnetization procedures were performed using different techniques.

Hysteresis loops experiments, using a translation inductometer within an electromagnet, (producing a field up to 1.6 T) revealed one behavior (Fig. 3(a)). The hysteresis after subtraction of the paramagnetic susceptibility shows open loops, indicating that the saturation was not reached. The coercive force (H_c) and the remanent coercive force (H_{cr}) are high, around 0.33 T and 0.5 T, respectively, indicating the presence of high coercive mineral(s), like hematite or goethite. The loop shows a slight wasp-wasted shape, which suggests a presence of a lower coercive force component. The wasp-wasted shape of the hysteresis loop obtained with brother samples thermally cleaned in air (Fig. 3(b)) show the presence of two magnetic phases (Roberts et al., 1995). The coercive forces are much smaller ($H_{cf} = 0.06$ T) in this case, indicating that heating resulted in the formation of a lower coercive force component, very likely magnetite. The isothermal remanent magnetization (IRM) acquisition curves are practically linear *and* did not reach the saturation (Fig. 4), indicating the presence of at least one high coercive force carrier of remanence, but there is no evidence of lower coercivity component effect on the remanence.

The Lowrie (1990) experiments have been applied using fields of 1.0 T (z axis), 0.4 T (y axis) and 0.15 T (x axis). They revealed two behaviors. The first corresponds only

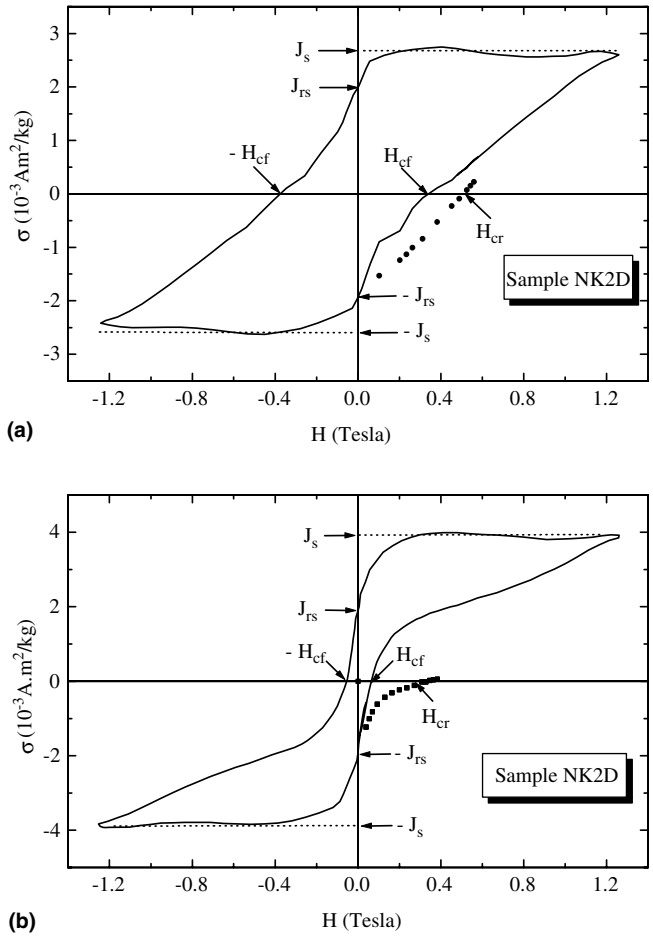


Fig. 3. Hysteresis loop of samples after subtraction of the paramagnetic susceptibility. H—induced field, H_{cr}—remanent coercive force, H_{cf}—coercive force of the ferromagnetic carriers, J—induced magnetization, J_{rs}—saturation remanent magnetization, J_s—saturation magnetization. (a) Untreated sample, (b) thermally cleaned sample.

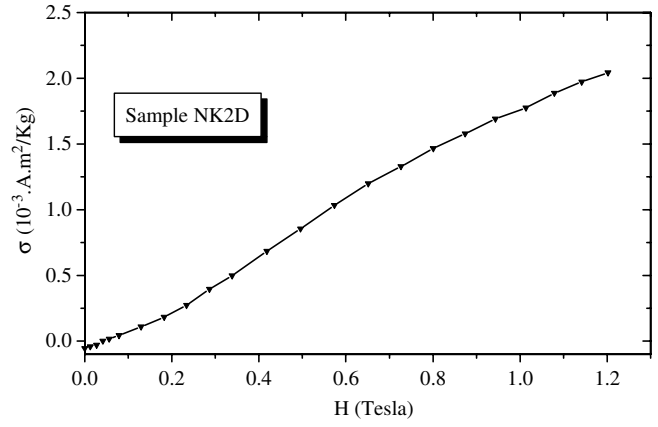


Fig. 4. IRM acquisition curve. The linear shape of the curve indicates the presence of at least one high coercive force carrier of remanence.

to site 1 and the second to the remaining sites. The thermal decay curves of the first behavior (Fig. 5(a)) confirmed the presence of hematite because of the high unblocking temperatures. However, the intensity of the high coercivity

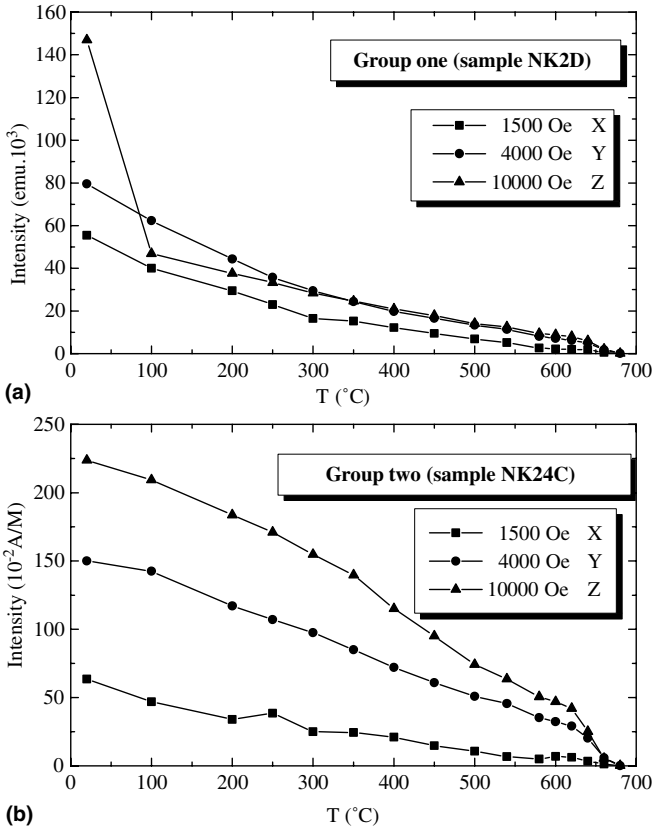


Fig. 5. Thermal cleaning of an IRM acquired by the application of an inductive field of 1.0 T along z axis, 0.4 T along y axis and 0.15 T along x axis. (a) The group one (site 1, sample NK2) shows presence of Goethite and hematite, (b) in the group two (all other sites) only hematite has been identified.

fraction (1.0 T) shows a major drop around 100 °C, suggesting the presence of goethite as a second high coercive force mineral. The curves of the two lower coercivity fractions (0.4 and 0.15 T) decrease monotonously with a slight slope change at temperatures around 300 °C suggesting the presence of another component in a small amount. For the second behavior (Fig. 5(b)), the main difference is the lack of drop of intensity at about 100 °C for the high coercive curve. Goethite was therefore probably not present in the sites other than the site 1.

The evolution of the natural remanent magnetization (NRM) intensity during the thermal treatments (Fig. 6) confirms, along with the high maximum blocking temperatures (about 670 °C), the presence of hematite. Some disturbances around 550–600 °C could be related to parasitic magnetization due to formation of magnetite. In addition, in samples of the site 1, the presence of goethite is also shown by drop of intensity.

The thermomagnetic analysis of magnetic susceptibility in low field $K(T)$ was achieved in air using KLY2 Susceptibility Bridge associated with high temperature CS2 furnace (Agico, Brno). The curves (Fig. 7(a) and (b)) show the presence of hematite, and the formation of magnetite at high temperature (non-reversible curves), particularly

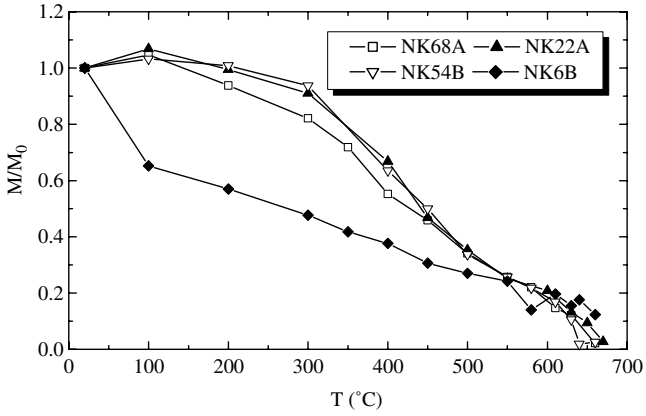


Fig. 6. Normalized curves (M/M_0) of the evolution of the magnetization intensity as a function of temperature T. The curve of sample NK6b corresponds to the first group, the other curves correspond to the second group.

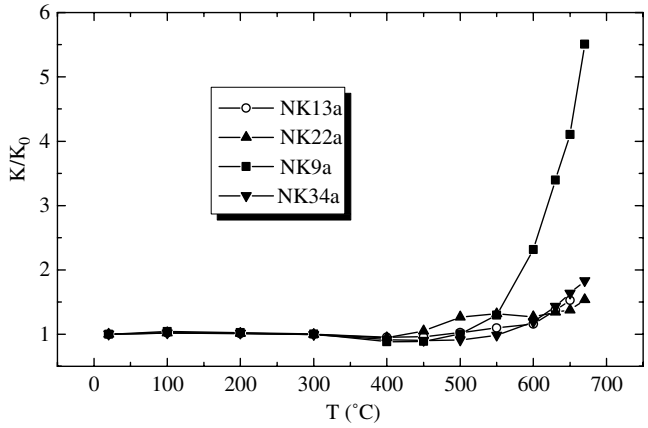


Fig. 8. Normalized curves (K/K_0) of the magnetic susceptibility at room temperature as a function of heating steps. The curve of sample NK9a corresponds to the first group (one), the other curves correspond to the second group (two).

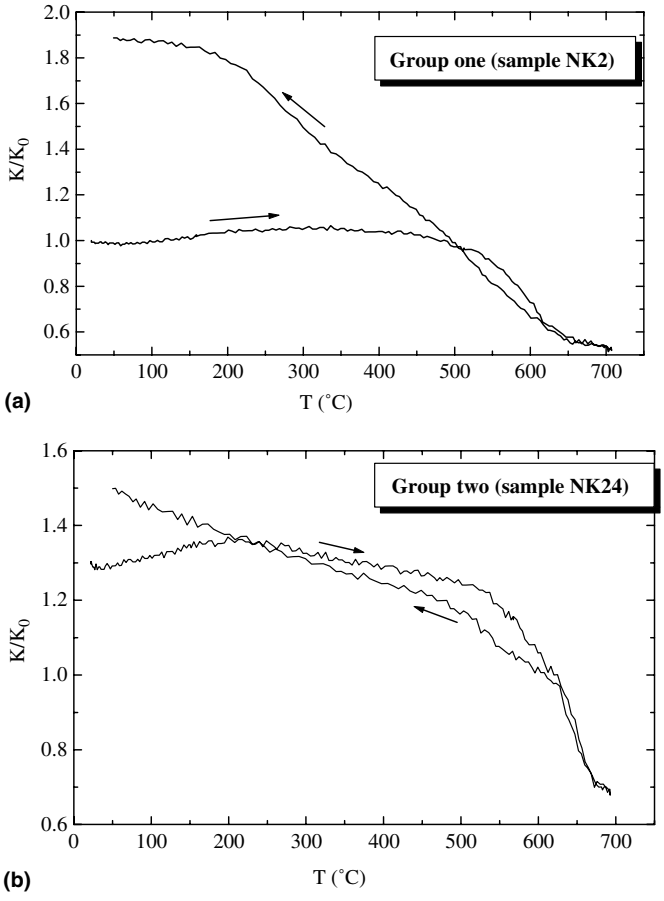


Fig. 7. Normalized thermomagnetic curves K/K_0 (susceptibility in low field as a function of temperature). (a) Group one (sample NK2); (b) group two (sample NK24).

in the sample from site 1 (Fig. 7(a)). Goethite does not give significant signal on the curve for sample from site 1.

Magnetic susceptibility measurements were done at room temperature using the KLY2 Bridge in order to monitor any mineralogical change. The curves (Fig. 8) show a very slight decrease of the susceptibility in a temperature interval 300–500 °C, followed by an increase of the susceptibility at temperatures higher than 500 °C. This increase is more important in samples of site 1, indicating the formation of magnetite in higher proportion.

As a conclusion, all these rock magnetic experiments clearly showed the high concentrations of hematite and the presence of small amount of goethite in site 1. Nedjari (1982) reported 7% of detrital hematite in the rocks. In addition, magnetite is formed during the thermal treatments with higher concentrations in the site 1.

3.3. Paleomagnetic methodology

Prior to any paleomagnetic analysis, all the samples were stored in a zero-magnetic field shielded space more than two weeks to reduce possible viscous remanent magnetization (VRM) acquired in situ or after the sampling. The NRM was measured using a JR-4 spinner magnetometer (Agico, Brno). Its intensity ranges between 112 and 673 mA/m with an average value of 369 mA/m.

The presence of hematite and goethite means that thermal demagnetization is the most appropriate to analyze the NRM. This study was carried out using thermal increments of 100 °C up to 300 °C, 50 °C up to 550 °C and 30 °C to 10 °C for higher temperatures. Demagnetization data for each sample were plotted on orthogonal vector plots (Wilson and Everitt, 1963; Zijderveld, 1967) and on equal-area projections. The directions of the linear segments were calculated using principal component analysis (Kirschvink, 1980). Data with MAD larger than 4° were systematically not retained (MAD is mostly lower than 3°). Mean directions data were determined using Fisher (1953) and bivariate (Le Goff, 1990; Le Goff et al., 1992) statistics. The paleomagnetic fold tests associated with progressive unfolding (McFadden, 1990; Watson and Enkin, 1993; Henry and Le Goff, 1994; Tauxe and Watson, 1994) were

used to assess the age of the characteristic remanent component relative to folding.

3.4. Paleomagnetic analysis

NRM directions of the Nekheila formation samples are spread along a girdle from southeast to the present day field direction. Therefore NRM probably contains components of recent viscous (VRM) or chemical (CRM) remagnetizations. Nevertheless, the distribution of NRM directions mainly in southeast quadrant suggests the presence of an ancient component.

A total of 117 samples had been thus subjected to progressive thermal demagnetization. These analyses revealed two different demagnetization behaviors.

The first behavior identified only in the first site, shows (Fig. 9(a)) two well-defined magnetic components. A low unblocking temperature component is isolated in all samples at temperatures below 200 °C and accounts for 40–80% of the initial NRM intensity. This component, whose direction is close to the present-day dipole field in the studied area, is a recent overprint, probably carried by goethite. The second component is a ChRM defined in all samples from temperature higher than 300 °C. However, above 580 °C, the obtained directions for the successive thermal steps are more or less randomly distributed. This is probably due to parasitic magnetization related to the formation of magnetite. This ChRM has a moderate positive inclination and a southeast declination.

The second demagnetization behavior concerns all the other samples. Orthogonal plot (Fig. 9(b)) shows first a weak unblocking temperature component. This component, with direction very close to that of the present day Earth magnetic field, is very likely of viscous origin. The ChRM with high unblocking temperatures (up to 680 °C) is isolated from 300 °C and has the same direction as in the site 1. Because of the weakness of mineralogical transformations during thermal treatments in these sites, the direction of the ChRM remains stable even for temperatures higher than 580 °C.

The ChRM, carried by hematite, has a mean direction defined by $N = 7$ sites, $D = 128.4°$, $I = 26.4°$, $k = 186$, $\alpha_{95} = 4.4°$ and $D = 131.8°$, $I = 15.7°$, $k = 196$, $\alpha_{95} = 4.3°$, respectively before and after dip correction. This mean direction, calculated by giving unit weight to each sample, is practically the same as those obtained while assigning the unit of weight to each site (Table 1). All ChRM directions are very well clustered with weak values of the ratio k_y/k_x: 1.6 and 1.1 before and after dip correction, respectively (characterizing both a weak ellipticity of the confidence zone—Le Goff, 1990 and Le Goff et al., 1992). Table 1 shows a good directions consistency for this ChRM within each site as reflected by the values of α_{95}, ranging from 2.1° to 4.5°. However, there is no significant improvement in the gathering of the site mean directions after dip correction (Fig. 10(a) and (b)).

Fold tests have been applied to the seven mean site directions to constrain the age of the ChRM relative to

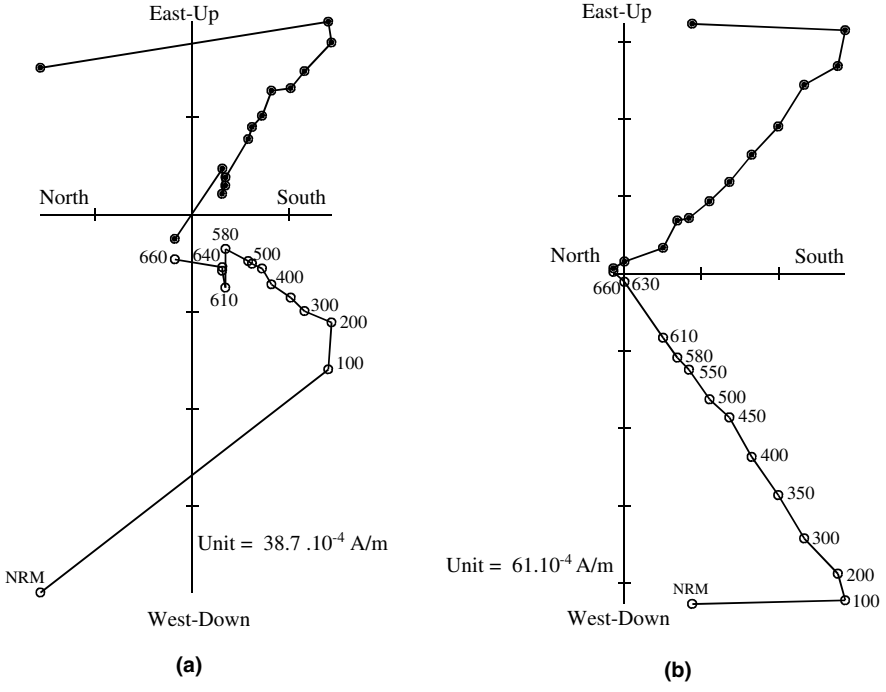


Fig. 9. Orthogonal vector plot or Zijderveld (1967) diagrams in geographical co-ordinates (filled and open symbols correspond respectively to horizontal and vertical plane). (a) Group one (sample NK8a). (b) Group two (sample NK12a).

Table 1
Mean characteristic directions (D, I) for the Autunian redbeds of Nekheila (Mezarif) before and after dip correction

Site	N	Before dip correction				After dip correction				Pole co-ordinates			
		D (deg)	I (deg)	k	α_{95} (deg)	D (deg)	I (deg)	k	α_{95} (deg)	La (°S)	Lo (°E)	K	A_{95} (deg)
Nekheila formation													
1	8	124.4	27.1	123	4.5	131.1	16.2	123	4.5	28.7	56.7		
2	19	126.5	32.5	227	2.1	134.7	20.8	227	2.1	29.5	51.8		
3	15	134.2	17.7	269	2.2	134.3	7.7	269	2.2	34.0	58.0		
4	26	131.3	26.7	170	2.1	131.6	16.7	170	2.1	28.8	56.1		
5	16	130.4	29.8	243	2.2	137.3	17.2	243	2.2	32.7	51.3		
6	17	127.4	22.5	147	2.8	128.1	12.7	147	2.8	27.7	60.5		
7	16	124.4	28.2	258	2.2	125.5	18.5	258	2.2	23.8	59.8		
Mean direction of samples													
	117	128.9	26.6	108	1.2	131.8	16.0	110	1.2	29.3	56.3		
	117									29.3	56.2	160	1.0
Mean direction of sites													
	7	128.4	26.4	186	4.4	131.8	15.7	196	4.3	29.4	56.4		
	7									29.3	56.4	322	3.4

(k, α_{95}, K, A_{95}) are Fisher's (1953) statistics parameters and (La, Lo) are respectively the latitude and longitude of the virtual south geomagnetic pole.

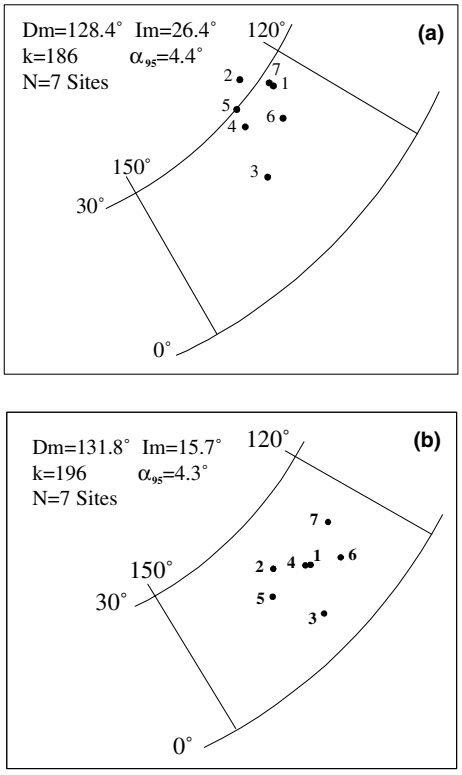


Fig. 10. The seven sites mean directions (1–7) of the ChRM of the Autunian Nekheila formation on an equal projection plot. Full symbols (circles) represent projections on the lower hemisphere. (a) and (b) respectively before and after dip correction.

the folding. The simple McElhinny (1964) fold test is not conclusive; the precision parameter increasing only by a factor of 1.1 (from 186 to 196) during progressive unfolding. We then tried several more relevant fold tests (McFadden, 1990; Tauxe and Watson, 1994; Watson and Enkin, 1993). For the McFadden test, the values of the parameter ξ_2 McFadden (1990) are always under the critical values at 95% (3.086) during progressive unfolding (1.852 and 1.251 before and after dip correction, respectively). The test is then undetermined. The result is the same with the two other tests, the values for 0% and 100% unfolding being included in the window of possible optimal unfolding.

Although, the fold tests cannot definitively solve the question about the age of this ChRM relative to tectonic events, some arguments plead in favor of the primary character of this ChRM. According to Nedjari (1982), hematite should be of detrital origin. At 100% unfolding compared to 0% unfolding, ellipticity k_y/k_x is weaker and precision parameters k and $k\delta$ larger. The direction of this ChRM leads to a paleomagnetic pole located at 29.3°S and 56.4°E ($K = 311$ and $A_{95} = 3.0°$). This pole is situated practically at the same position as that from the neighboring Autunian Abadla formation (Merabet et al., 1998). It is also very close to Morel et al.'s (1981) pole obtained in the upper non-dated part of the Abadla formation.

4. Regional fold test

The Mezarif basin is geologically the symmetric of the large Abadla basin relative to the Zousfana anticline axis, and the overall area was moderately folded during the late Hercynian orogeny. Since a paleomagnetic study had been performed in the equivalent Autunian red formation of Abadla (Merabet et al., 1998), application of the fold test to combined Abadla and Nekheila paleomagnetic data was the final aim of this study.

The combined ChRM mean directions (Table 2) are coherent between samples ($N = 213$) and sites ($N = 18$). The mean direction for the 18 sites is $D = 128.4°$, $I = 20.9°$, $k = 101$, $\alpha_{95} = 3.3°$ and $D = 130.8°$, $I = 14.0°$,

Table 2
Data from Nekheila (see Table 1) and Abadla (Merabet et al., 1998) formations (D and I of site 3 and 11 of Abadla basin were determined respectively from 8 great circles and 12 great circles and 7 ChRMs)

Site	N	Before dip correction				After dip correction				Pole co-ordinates			
		D (deg)	I (deg)	k	α_{95} (deg)	D (deg)	I (deg)	k	α_{95} (deg)	La (°S)	Lo (°E)	K	A_{95} (deg)
Abadla formation													
1	8	127.1	23.4	107	4.8	128.9	18.2	107	4.5				
2	4	127.2	27.4	198	5.0	129.4	22.2	198	5.0				
3	8	133.9	8.0	94	5.1	134.3	2.3	94	5.1				
5	4	125.0	25.0	106	6.8	128.1	19.0	106	6.8				
6	8	129.3	20.0	90	5.2	131.5	13.1	90	5.2				
8	12	124.5	12.1	187	3.0	125.8	8.3	187	3.0				
9	10	135.3	12.8	191	3.2	136.5	9.4	189	3.2				
10	12	126.4	14.0	226	2.7	128.0	11.2	226	2.7				
11	19	125.9	8.7	195	2.3	126.9	6.0	195	2.3				
12	15	129.2	21.6	118	3.3	131.7	17.8	118	3.3				
13	16	127.7	18.3	123	3.2	129.9	14.8	123	3.2				
Nekheila and Abadla formations													
Mean direction of samples													
	213	128.3	22.5	74	1.1	130.8	14.8	98	1.1	29.0	56.9		
	213									29.0	57.0	149	0.8
Mean direction of sites													
	18	128.4	20.9	101	3.3	130.8	14.0	155	2.6	29.3	57.3		
	18									29.2	57.2	330	1.8

For legends see Table 1.

$k = 155$, $\alpha_{95} = 2.6°$, respectively before and after dip correction. This mean direction is practically the same as that obtained from sample-mean directions (Table 2). The directions of the combined ChRMs are very well clustered with weak values of the ratio k_y/k_x: 2.1 and 1.1 before and after dip correction, respectively (Le Goff, 1990; Le Goff et al., 1992). The gathering of site mean directions of these combined ChRMs is significantly improved after dip correction (Fig. 11(a) and (b)).

During progressive unfolding for 18 sites, this ellipticity is minimum for about 100% unfolding (indicating Fisherian distribution). On the contrary, value of 2.1 for 0% unfolding is significant of a non-Fisherian distribution. Different fold tests (McElhinny, 1964; McFadden, 1990; Watson and Enkin, 1993; Tauxe and Watson, 1994) have been applied to combined data from the two basins. The McElhinny (1964) test is here again inconclusive, with k_2/k_1 (ratio before and after unfolding) equal to 1.53 or with $\sqrt{k_x \cdot k_y}$ parameter (Henry and Le Goff, 1994): values before and after unfolding equal to 150 and 187, respectively. However, the values of these two parameters are higher for 100% unfolding than for 0% unfolding (Fig. 12). The McFadden (1990) fold test at 99% and 95% leads to critical values of 6.919 and 4.940, respectively. Measured ξ_2 value for 0% unfolding (7.593) is greater than both critical values (6.919 and 4.940). On the contrary, for 100% unfolding the value (1.458) is smaller than that critical values. This result clearly points out that the ChRM is pre-folding. In addition, during progressive unfolding (Fig. 13), the value of ξ_2 is minimum close to 100% unfolding. The Tauxe and Watson (1994) and Watson and Enkin (1993) tests yields the same positive fold test, the optimal

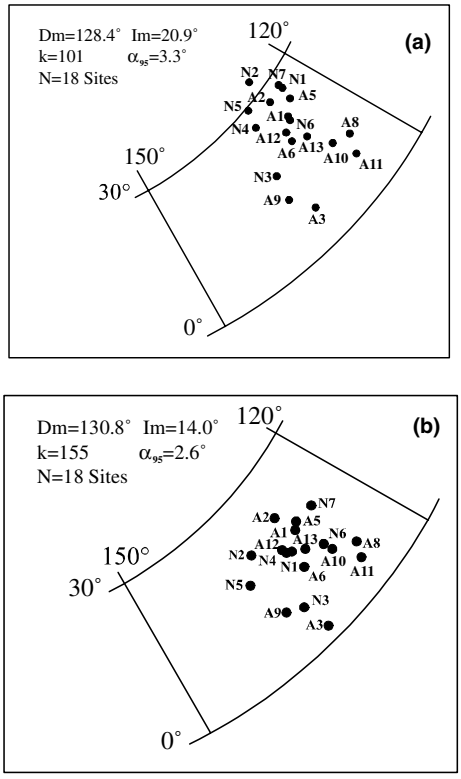


Fig. 11. The 18 sites mean directions of the combined ChRMs of Nekheila (N1-7) and Abadla (A1-13; Merabet et al., 1998) before (a) and after (b) dip correction. Full symbols (circles) represent projections on the lower hemisphere.

unfolding being between 87% and 160% unfolding and between 100% and 160% unfolding, respectively. Though mean optimal unfolding happens at values higher than

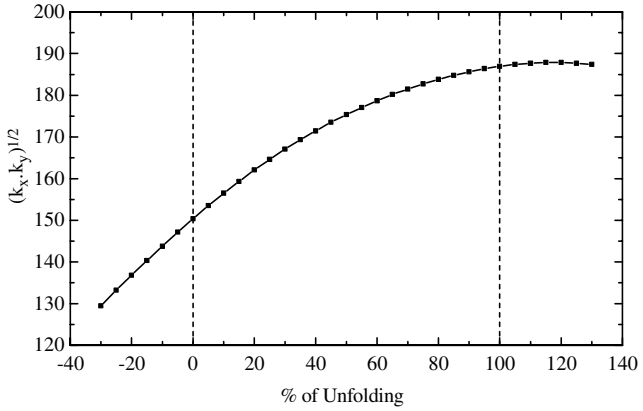


Fig. 12. Variation of $\sqrt{k_x \cdot k_y}$ parameter (Henry and Le Goff, 1994) during progressive unfolding.

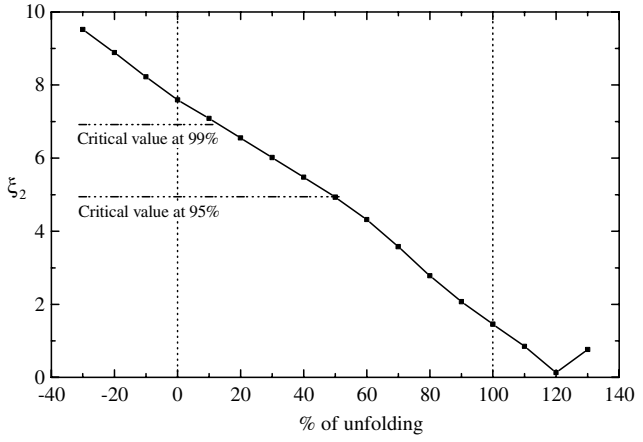


Fig. 13. Variation of ξ_2 (McFadden, 1990) during progressive unfolding.

100%, this latter value is always included within the optimal unfolding window and the fold tests are positive.

Thus, the mean direction of the combined Abadla and Nekheila paleomagnetic data leads to a paleomagnetic pole situated at 29.2°S and 57.2°E ($K = 330$ and $A_{95} = 1.8°$; Table 2) with an early age well constrained by the positive fold test. The folding is in fact attributed to a late Hercynian phase, during or just after the Autunian, and the magnetization was acquired before the folding.

5. Discussion and conclusion

From a geodynamical point of view, some recent results seem to strengthen the suggestion of an eastward drift from South Africa of the apparent pole during the Upper Devonian (Henry et al., 1992). Indeed, a new Upper Devonian datum (Derder et al., 2003; Ouabadi et al., 2004) confirms the location of the Upper Devonian APWP in Southern Africa (Aifa et al., 1989) and allows the choice between different possible ages proposed by Smith et al. (1994) for the Hazzel Mati pole. This later then corresponds very likely to a remagnetization.

Besides, the new Lower Serpukhovian to Lower Moscovian datum (Derder et al., 2001d) is in an intermediate location between the Upper Devonian and the other Upper Carboniferous poles. Indeed this datum has been acquired in a probably slightly older formation. This assertion is corroborated by the presence of both polarities contrary to previous studies of Mid Carboniferous Saharan formations where only reversed polarities were isolated. The counter clockwise rotation of Africa during Hercynian times (Matte, 1986; Henry et al., 1992) was therefore not finished during the Serpukhovian period (Fig. 14).

The results obtained in the well-dated continental Autunian formation at Nekheila show the existence (besides a recent viscous or remagnetization event) of a ChRM that does not pass the fold test due to presence of a unique limb. However, the corresponding paleomagnetic pole (29.3°S, 56.4°E, $K = 311$, $A_{95} = 3.0°$) is very close to that (29.1°S, 57.8°E, $K = 462$, $A_{95} = 2.0°$) of the Autunian red formation of the neighboring Abadla basin. As the Abadla and Mezarif basins are situated on both sides of the large Zouzfana anticline area, fold test had been applied to ChRMs of both Autunian units. The positive result of this test now constrains similarly the early age of the magnetization in these units. The paleomagnetic pole (29.2°S, 57.2°E, $K = 330$, $A_{95} = 1.8°$) corresponding to combined sites of the two Autunian units is thus the sole African pole with an Autunian age constrained by a positive fold test. The other already published African poles of the Stephano–Autunian formations in the Illizi (Derder et al., 1994) and Tindouf (Henry et al., 1999) basins are close to that of the Abadla–Mezarif pole, and their Stephano–Autunian age is then also confirmed. The African APWP has

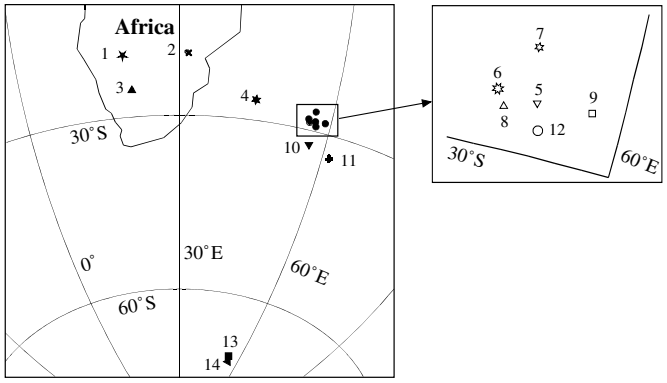


Fig. 14. Paleomagnetic poles of the Algerian Saharan stable areas from Devonian to Lias. 1: Famenian (Aifa et al., 1989); 2: Famenian–Franian (Derder et al., 2003); 3: Tournaisian (Aifa et al., 1989); 4: Lower Serpukhovian to Lower Moscovian (Derder et al., 2001d); 5: Namurian (Merabet et al., 1999); 6: Bashkirian (Derder et al., 2001b); 7: Upper Namurian–Lower Moscovian (Daly and Irving, 1983); 8: Moscovian (Henry et al., 1992); 9: Moscovian (Derder et al., 2001c); 10: Lower-Stephanian (Henry et al., 1999); 11: Stephano–Autunian (Derder et al., 1994), 12: Autunian (this study); 13: Late Triassic–Liassic (Kies et al., 1995); 14: Liassic (Derder et al., 2001b).

therefore an Upper Carboniferous–Lower Permian part well specified and constrained in age.

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